Power BI Tool

Power BI Desktop is used for authoring and designing reports and dashboards. If you want to share your reports and dashboards with others or collaborate with a team, you'll need to publish your work to the Power BI Service, which is a cloud-based platform.

Why Do We Need Power BI?

In today's world, data is the new oil. Organizations need a tool that can help them to understand the large amount of data that they are collecting. Microsoft Power BI is introduced to fulfill their needs, It is a powerful data visualization and analysis tool that allows businesses to turn raw data into actionable insights and reports.

Power BI helps users who do not come from an analytical background. Still, they can create reports on the Power BI desktop.

Features of Power BI

Some most important Power BI features are given below :

- **Data Visualization**: Power BI is used by organizations to create interactive reports and dashboards that visualize data in various ways, including charts, graphs, and maps.
- **Collaboration**: Power BI allows multiple users to work on the same report or dashboard at the same time. This makes it easy for teams to collaborate, share data and share insights.
- **Data Transformation:** Power BI can transform raw data into useful information that is a more usable format by cleaning and manipulating data. This allows businesses to work with clean, accurate data that is easy to analyze.
- **Integration**: Power BI integrates with a wide range of other Microsoft products or applications like Excel, SharePoint, and Teams. It helps in the workflow of the projects.
- **APIs for integration**. This feature provides developers with sample code and APIs for embedding the Power BI dashboard in other software products.

Power BI Architecture

The whole process of data sourcing to the creation of reports and dashboards consists of four basic steps. Many programs and processes work together to get the required results.

1. **Sourcing Data –** Power BI has a large range of data sources. Data can be imported from cloud-based online sources, and files in your system. There is a limit of 1 GB on importing data from online services. Power BI

data sources are Excel, Text/CSV, XML, JSON, Oracle Database, Azure SQL Database, etc.

2. **Transforming the Data –** Cleaning and pre-processing are done before visualizing the data. It means removing missing values and useless data from the rows and columns. There are some rules to transforming and loading the datasets into the warehouse.
3. **Report and Publish –** Now, the report is created based on requirements, after cleaning and transforming the data. A report is data visualization with different filters and constraints represented in the form of graphs, pie, charts, etc.
4. **Creating Dashboards –** Power BI dashboards are created by pinning independent elements of live reports. It is created after the publishing of the report to the BI service. After reports get saved, the visual maintains the filter settings.

Power BI Architecture Components

- **Power Query:** Power Query is included in the Power Query Editor of Power BI Desktop, and It allows users to connect distinct information from multiple sources and convert them to satisfy their business requirements
- **Power BI service:** Power BI Service connects Power View, Power Pivot, Power BI Report Server, Power Q&A, and other components with the Workspace and allows you to connect with the data.
- **Power Pivot:** Power Pivot is a data modeling technique that uses the Data Analysis Expression (DAX) language to create data models.
- **Power View:** Power View create graphs, maps, charts, and other visuals with drag-and-drop features. Power View can connect and filter different data sources to make a report on a single device.
- **Power Map:** Power Map allows you to base on the country, longitude, and latitude, Bing Maps shows the exact geospatial visuals of complex business information.
- **Power BI Desktop:** Power BI Desktop brings everything on a single platform, including Power Pivot, Power Query, Power View, etc. It is free software that you can install on a PC or laptop to create dashboards and reports.
- **Power BI App:** Power BI mobile applications are available for iOS, Android devices, and Windows. It does not matter whether your data is stored on the cloud or on-premises servers, you can access reports, run queries, share reports, and also get personalized notifications.
- **Power Q&A:** Power Q&A uses Natural Language Processing and integrates it with Cortana (Microsoft's virtual assistant) to get the answers to your query.

Advantages of Power BI

- **Cost Savings or Affordability**: Power BI is a cost-effective solution for businesses, as it allows them to make the most of the data they have without investing in other expensive analytics software. The Power BI

desktop version is free, we can download it and start making data models and reports.
- **Better Data Insights:** Power BI helps businesses make sense of their data, allowing them to identify trends, track performance, and make data-driven decisions.
- **Improved Collaboration**: Power BI makes it easy for teams to work together, share insights, and collaborate on reports and dashboards.

Installation of Power BI

To install Power BI, follow these steps:

1. Check System Requirements:

   Before installing Power BI, ensure that your computer meets the system requirements. You can find the latest system requirements on the official Microsoft Power BI website.

2. Download Power BI Desktop:

   Power BI Desktop is the application used for creating reports and dashboards. You can download it for free from the official Power BI website.

   Go to the official Power BI Desktop download page (https://powerbi.microsoft.com/en-us/desktop/).

   Click the "Download free" button.

3. Run the Installer:

After the download is complete, run the installer by double-clicking on the downloaded file.

4. Installation Process:

   Follow the installation prompts and instructions. You can usually leave most of the default settings as they are, but make sure to review the terms and conditions and privacy settings.

5. Launch Power BI Desktop:

   Once the installation is complete, you can launch Power BI Desktop.

Preparation of Dashboard with Sample Data

Creating a dashboard for a random report in Power BI involves several steps. Here are sample instructions to create a simple dashboard using random data for demonstration purposes:

Step 1: Data Preparation

1. *Generate Random Data:*
   - *For this sample, let's generate random data in an Excel spreadsheet. Create columns for "Product," "Sales," "Year," and "Region." Populate the spreadsheet with fictional data.*

Step 2: Data Import

1. *Open Power BI Desktop:*
   - *Launch Power BI Desktop on your computer.*
2. *Get Data:*
   - *In Power BI Desktop, click on "Get Data" to import data.*
3. *Select Data Source:*
   - *Choose "Excel" or the appropriate data source for your random data file.*
4. *Load Data:*
   - *Select the Excel file you created in Step 1 and load the data into Power BI.*

Step 3: Data Modeling

1. *Data Modeling View:*
   - *In Power BI Desktop, navigate to the "Model" view.*
2. *Manage Relationships:*
   - *If applicable, create relationships between tables. In this sample, you may not need to establish relationships as you're using a simple Excel file.*

Step 4: Data Visualization

1. *Report View:*
   - *Switch to the "Report" view in Power BI Desktop.*
2. *Create Visuals:*
   - *Create visuals such as a bar chart to represent "Sales by Product," a line chart for "Sales by Year," and a map to display "Sales by Region."*
3. *Customize Visuals:*
   - *Customize your visuals by adjusting titles, colors, and labels. You can also add slicers to allow filtering by year or region.*

Step 5: Dashboard Creation

1. *Dashboard View:*
   - *Go to the "Dashboard" view in Power BI Desktop.*
2. *Create New Dashboard:*

- *Create a new dashboard by clicking "New Dashboard" and providing a name, such as "Sample Sales Dashboard."*

3. *Add Visuals:*
    - *Drag and drop the visuals you created in the "Report" view onto the dashboard canvas. Arrange them to create an appealing layout.*
4. *Add Text and Images:*
    - *Enhance the dashboard by adding text boxes with descriptions and images for context.*

Step 6: Interaction and Filtering

1. *Set Up Interactions:*
Configure interactions between visuals to allow cross-filtering. For example, selecting a product in the bar chart should update the line chart and map accordingly.

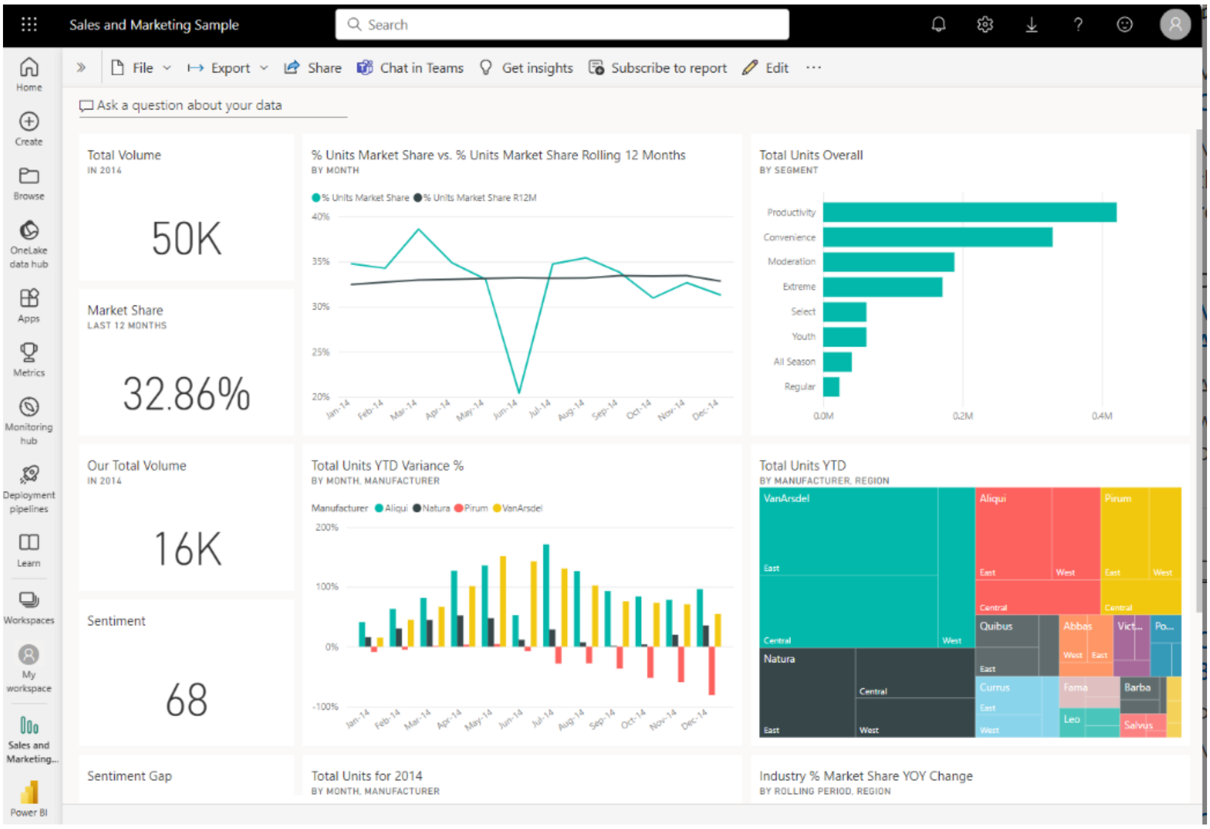
Step 7: Publish and Share

1. *Save the Report:*
    - *Save your Power BI Desktop report.*
2. *Publish to Power BI Service:*
    - *Publish your report to the Power BI Service to make it available online.*
3. *Share with Others:*
    - *Share the dashboard with colleagues or stakeholders, granting them appropriate permissions to view or interact with the report.*

Step 8: Test and Refine

1. *Testing and Feedback:*
    - *Test the dashboard and gather feedback from users.*
2. *Refine as Needed:*
    - *Make adjustments to the dashboard layout, visuals, and interactions based on user feedback and changing business requirements.*

How Dashboard Data Report Visualize:



DIFFERENCE BETWEEN POWER BI AND TABLEAU

Power BI and Tableau are both popular business intelligence and data visualization tools, but they have differences in terms of features, pricing, integration capabilities, and user experience.

Features	**POWER BI**	**Tableau**
Company	Microsoft	Salesforce
Ecosystem	It is tightly integrated with the Microsoft ecosystem, including Azure services and Office 365.	It has a strong focus on data exploration and interactive visualizations.
Ease of Use	User-friendly interface and integration with Excel	Although powerful and flexible interface but some users might find its learning curve slightly steeper than Power BI.
Data Source Connectivity	Similarity: Both tools support a wide range of data sources, including databases, cloud services, spreadsheets, and more.	
Data Source Connectivity	Its seamless integration with various Microsoft services like Azure SQL Database and Excel, making it a good	Tableau also supports diverse data sources and has connectors for various databases, cloud platforms, and web services.

	choice for organizations heavily invested in the Microsoft ecosystem	
Data Transformation and Modeling	Its data transformation capabilities through Power Query, which allows users to clean and shape data before visualization.	It has data preparation tools like "Tableau Prep" that help users clean and reshape data for analysis.
Visualization Capabilities	Tableau is often praised for its interactive and dynamic visualizations that allow users to drill down and explore data in-depth.	It also try to improved its visualization capabilities over time, offering features like custom visuals and drill-through actions
Pricing	- Power BI offers both a free version with limited features and several paid plans, including Power BI Pro and Power BI Premium. The pricing is typically user-based.	Tableau's pricing model can be more complex, with options like Tableau Desktop (for authoring) and Tableau Server (for sharing) available. The cost can increase as the number of users and required features grow.
License		**Tableau Desktop-per user basis**
Collaboration and Sharing	Power BI allows collaboration through its cloud-based service, Power BI Service. Users can publish reports and dashboards, share them with colleagues, and collaborate in real-time.	Tableau offers collaboration through Tableau Server or Tableau Online, where users can share and collaborate on visualizations.
Mobile Experience	Both tools offer mobile apps that allow users to view and interact with reports and dashboards on mobile devices.	
Decision Remark	To purchase software depends on factors like our organization's existing technology stack, budget, preferred user experience, and the specific features and capabilities you require for your data analysis and visualization needs.	

Additional Information Power BI Data Visualization Tools:

Power BI	Use
Desktop	It is free and use to create reports and visualization
Pro	This license is user-based and allows users to share and collaborate on reports and dashboards using the Power BI Service. It usually requires a monthly or annual subscription fee per user.
Premium	This license offers dedicated capacity for better performance, larger datasets, and additional features. It's suitable for organizations with a higher demand for data analytics. Pricing is based on capacity and user requirements

Additional Information Tableau Data Visualization Tools:

Tableau	Use
Desktop	This is the authoring and publishing tool for creating visualizations. It's licensed on a per-user basis.
Server	This allows sharing and collaborating on visualizations within an organization. It can be licensed on a per-user or per-core basis, with costs varying based on the number of users and cores.
Online	This is a cloud-based version of Tableau Server. It's also licensed on a subscription basis.
Reader:	Tableau This is a free application that allows users to view and interact with Tableau visualizations created by others.